

Urban Drift

2D Traffic Survival Game

Manage speed, read traffic, survive to the finish line.

Developer

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Introduction

For this project, we're planning to build a small 2D game based on everyday traffic situations. The core idea is simple — take something everyone experiences and turn it into a fun, interactive challenge. Rather than a racing game, the focus will be on survival and quick decision-making as traffic gets progressively harder to navigate.

We're aiming to keep the scope manageable: clean 2D visuals, straightforward mechanics, and traffic behavior that feels natural enough to be engaging. The goal is to create something small but polished — a short experience that's easy to pick up and actually enjoyable to play.

Through this project, we want to explore basic game development concepts like control, movement, and gameplay design. It will also help us understand how simple mechanics can create engaging gameplay. Since this is our first time doing anything like this, we want to try our best in making it successful.

Inspiration

The inspiration for our game came from our interest in car-based games, especially titles like Need for Speed, which we have played for a long time. While most of these games focus on high-speed racing, we wanted to explore a different idea involving cars but without traditional racing mechanics. This led us to think creatively about how vehicles could be used in a more strategic or traffic-based environment. After brainstorming together, we decided to design a small traffic management-style game where the challenge comes from control, timing, and decision-making rather than speed. This approach allowed us to create something unique while still building on a concept we enjoy. Since this is our first project, we were motivated to make something simple yet engaging that reflects both our creativity and learning process.

Game Overview



Genre

2D top-down survival game



Platform

PC build



Objective

Drive from start to finish without collisions. Survive, finish fast, score higher



Gameplay

Top-down roads, dynamic traffic streams, velocity management

Gameplay Mechanics

01

Movement System

Direct 2D control — W/A/S/D for precise lateral and longitudinal movement within lane constraints.

03

Collision System

Any vehicle-vehicle collision ends the run.

02

Velocity Control

Up Arrow to increase speed, Down Arrow to decrease. Speed affects braking distance and collision risk.

04

Survival Mechanics

Manage gaps, anticipate automatic moves, and adjust velocity to avoid chain collisions and reach finish alive.

Controls

Movement:

W = forward/nudge up lane;

S = brake/reverse slightly

A/D = steer left / right within road lanes.

Speed Management:

Up Arrow = Increase velocity (hold for continuous acceleration).

Down Arrow = Decrease velocity (tactical slow down, safer gaps).

Design note:

Velocity is analog (not instant) . Momentum and stopping distance matter.

Vehicle Types & Behavior



Private Cars

Fast acceleration, nimble lane changes.



Rickshaws

Unpredictable lateral shifts, lower top speed, frequent slow-downs, creates cluttered lanes.



Trucks

Slow, large collision radius, limited maneuverability; require early planning to pass safely.



CNGs

Moderate speed, tendency to drift between lanes, variable acceleration.



Trains

High speed on fixed tracks, long stopping distance, massive collision impact, and limited directional control; requires precise timing and careful crossing management.

Game Sample Idea





Level Design (5 Levels)

Level 1 — Tutorial: An introductory level for the players to get used to the game.

Level 2 — Easy level

Level 3 — Medium

Level 4 — Hard

Level 5 — Extreme

Scoring System



Time-Based Score

Primary score scales inversely with time to finish — faster runs yield higher base points.



Safe Driving Bonus

Points awarded for clean segments (no close calls, no near-misses). Conserving speed earns multipliers.



Risk-Reward

Overtaking at high speed grants bonus points but raises collision risk — core tension in gameplay.



PERSONAL BEST LOG

Tracks the player's highest scores and best performances, encouraging continuous improvement and replay

Key Game Features



2D Top-Down Graphics

Stylized, readable lanes with visual clarity.



Speed Management

Player-managed velocity is the primary risk-control mechanic.



Dynamic Traffic

Procedural vehicle spawn patterns create varied encounters each run.



Robust Collision Detection

Immediate game over on vehicle-vehicle collisions — encourages cautious play.

Further Plans

- Weather effects (rain, fog) to alter visibility and traction
- Expanded vehicle roster and level variety
- Upgraded visuals

Tools

1. C++ Compiler (MinGW, Clang, or MSVC).
2. VSCode with C/C++ Extension.
3. Raylib library installed or linked in the project.
4. [Freesound.org](https://freesound.org)(for sound sample)

Summary:

Urban Drift is a focused 2D top-down traffic survival prototype that challenges players to master speed control and situational awareness across five escalating levels.

What players learn: risk assessment, timing, and adaptive speed management under pressure.

Thank you.